massless particles, for example. Wrapped strings are different in this regard.)

How does the existence of wrapped string configurations affect the geometrical properties of the dimension around which the strings wind? The answer, first recognized in 1984 by the Japanese physicists Keiji Kikkawa and Masami Yamasaki, is bizarre and remarkable.

Let’s consider the last cataclysmic stages of our variant on the big crunch in the Garden-hose universe. As the radius of the circular dimension shrinks to the Planck length and, in the mold of general relativity, continues to shrink to yet smaller lengths, string theory insists upon a radical reinterpretation of what actually happens. String theory claims that all physical processes in the Garden-hose universe in which the radius of the circular dimension is shorter than the Planck length and is decreasing are absolutely identical to physical processes in which the circular dimension is longer than the Planck length and increasing! This means that as the circular dimension tries to collapse through the Planck length and head toward ever smaller size, its attempts are made futile by string theory, which turns the tables on geometry. String theory shows that this evolution can be rephrased—exactly reinterpreted—as the circular dimension shrinking down to the Planck length and then proceeding to expand. String theory rewrites the laws of short-distance geometry so that what previously appeared to be complete cosmic collapse is now seen to be a cosmic bounce. The circular dimension can shrink to the Planck-length. But because of the winding modes, attempts to shrink further actually result in expansion. Let’s see why.

The Spectrum of String States*

*Some of the ideas in this and the next few sections are rather subtle, so don’t be put off if you have trouble following every link in the explanatory chain—especially in a single reading.

The new possibility of wound-string configurations implies that the energy of a string in the Garden-hose universe comes from two sources: vibrational motion and winding energy. From the legacy of Kaluza and Klein, each depends on the geometry of the hose, that is, on the radius of its curled-up circular component, but with a distinctly stringy twist, since point particles cannot wrap around dimensions. Our first task, then, will be to determine precisely how the winding and